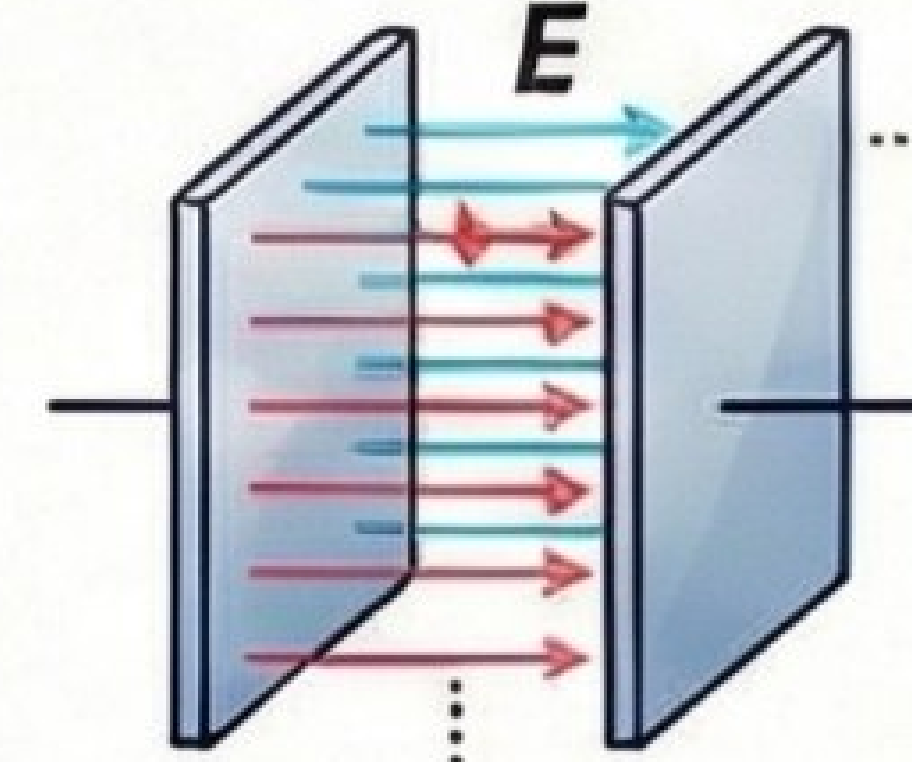


A Visual Guide to Electromagnetic Waves

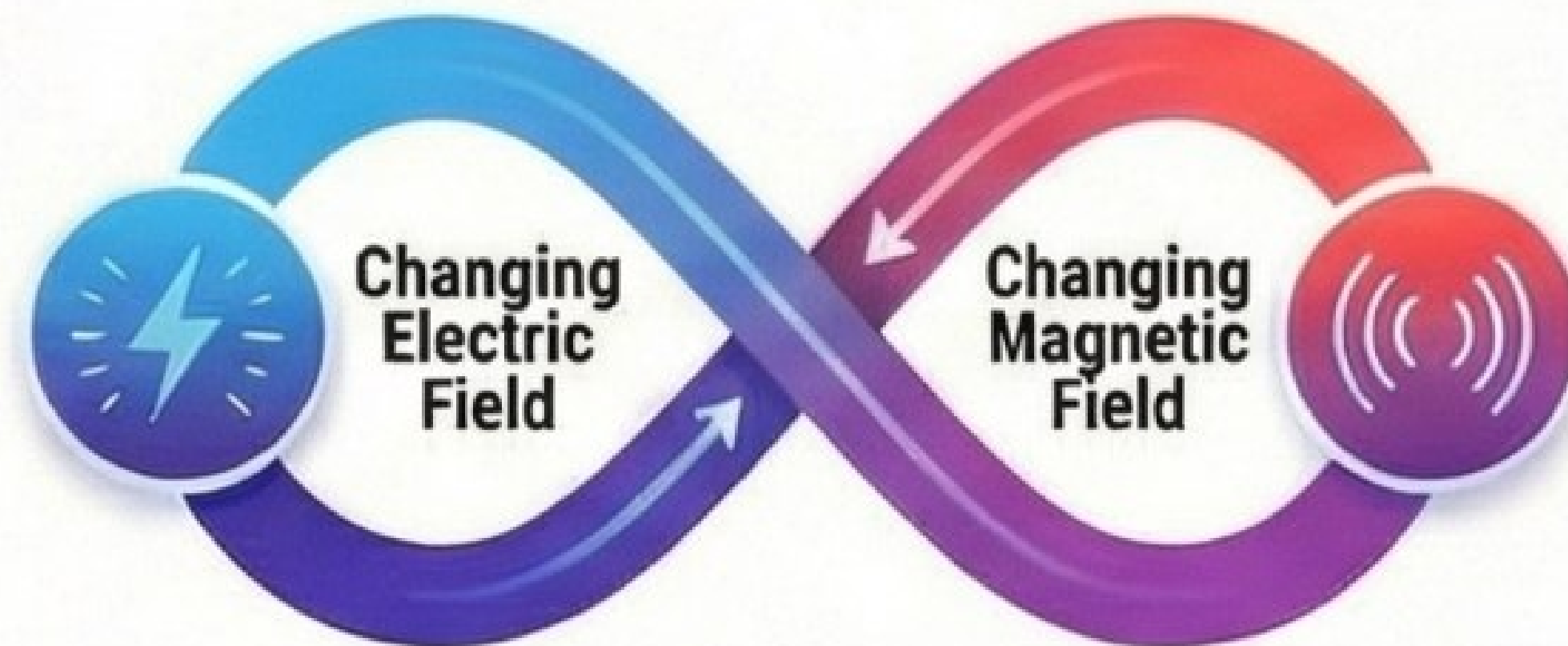
The Birth of an EM Wave



Changing Electric field Capacitor

The Missing Link: Displacement Current (Maxwell)
Discovered by Maxwell, this is the current produced by a changing electric field.

Changing Fields Create Each Other

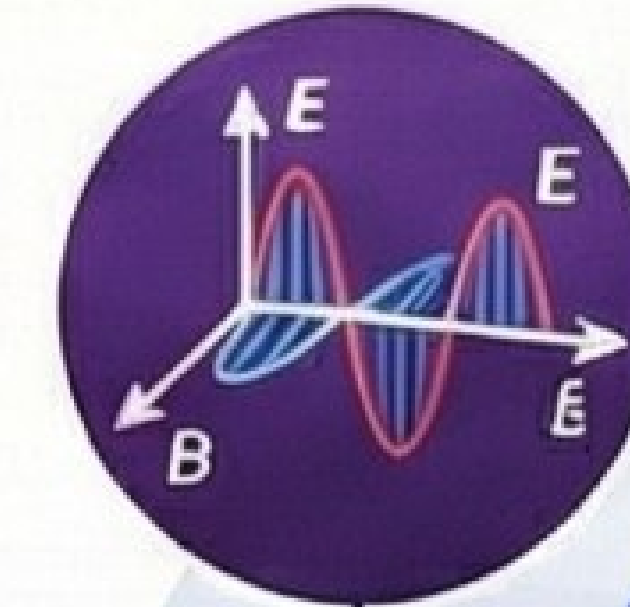


The Self-Perpetuating Cycle

A changing E-field creates a changing B-field, which in turn creates a new changing E-field. This cycle propagates through space, independent of the original source, turning on EM wave.

Core Properties of EM Waves

Transverse in Nature
The electric and magnetic fields oscillate perpendicular to each other and to the direction the wave is traveling.



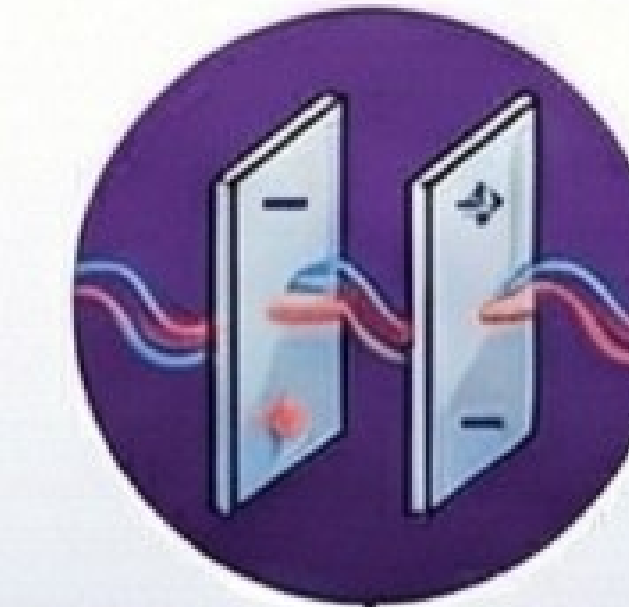
No Medium Required
Unlike sound, EM waves can travel through the vacuum of space.



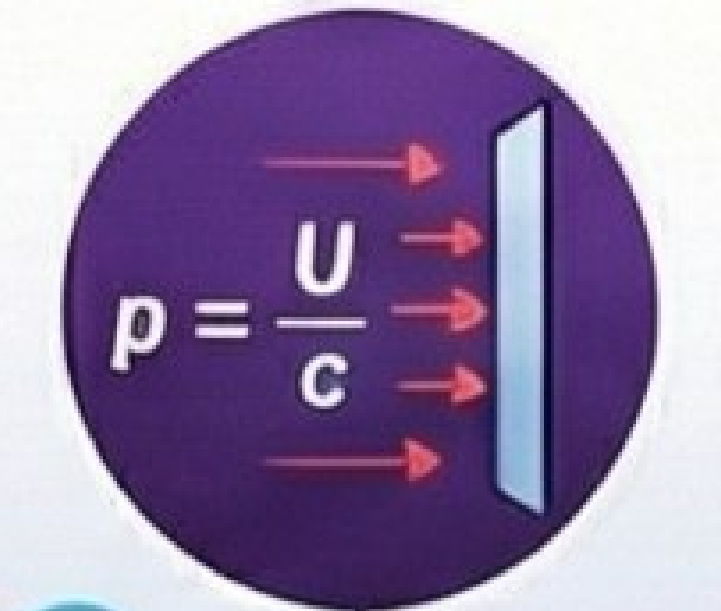
Travel at the Speed of Light
In a vacuum all EM waves travel at a constant speed, $c = 3 \times 10^8$ m/s.



Electrically Neutral
They have an energy and are therefore not deflected by electric or magnetic fields.



Carry Energy & Momentum
EM waves transport energy and can exert a force, known as radiation pressure, on a surface.
Momentum (p) = Energy (U) / c



Maxwell's 4 Unifying Equations

1. Gauss's Law for Electricity



$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

Defines the net electric flux through a closed surface in the Coulomb electric field, it confirms that single electric charges (monopoles) exist.

2. Gauss's Law for Magnetism



$$\oint \vec{B} \cdot d\vec{A} = 0$$

States that the net magnetic flux through any closed surface is zero. This implies that magnetic monopoles (isolated North or South poles) do not exist.

3. Faraday's Law of Induction



$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

A changing magnetic flux induces an electric field. This is the principle behind electric generators.

4. Ampere-Maxwell Law (Modified)



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(I_c + \epsilon_0 \frac{d\Phi_E}{dt} \right)$$

A magnetic field is generated by both a current in a wire (conduction current) and by a changing electric field (displacement current).

Everyday Physics: EM Waves in Action



Why do welders wear special glasses?
To protect their eyes from the intense ultraviolet (UV) radiation produced by the welding arc, which can cause serious eye damage.



How do microwaves heat food?
The oven produces microwaves at the resonant frequency of water molecules. This causes the water in the food to vibrate rapidly, generating heat.



Why do radio signals bend around buildings but light doesn't?
Diffraction (bending) occurs when a wave's wavelength is comparable to the size of an obstacle. Radio waves have long wavelengths (meters) similar to buildings, while light's wavelength is microscopic.

The Electromagnetic Spectrum

The Inverse Relationship:

Highest Frequency / Shortest Wavelength

As you move from Gamma rays to Radio waves across the spectrum, frequency and energy decrease, while wavelength increases.

Lowest Frequency / Longest Wavelength



Gamma Rays

Source:

Source: Radioactive Decay

Uses: Cancer therapy, sterilizing medical equipment, detecting cracks in engines



X-Rays

Source:

Source: Fast-moving electrons hitting a metal target

Medical imaging (bone fractures), airport security scanners



Ultraviolet (UV)

Source:

Source: The Sun, special lamps

Sterilization, food preservation, detecting forged currency, treating skin conditions



Visible Light

Source:

Source: The Sun, light bulbs

Vision, photography, fiber optics



Infrared (IR)

Source:

Source: Hot bodies, molecular vibrations

Thermal imaging, remote controls, physical therapy (heat lamps), night vision



Microwaves

Source:

Source: Magnetron tubes

Microwave ovens, radar systems, telecommunications, Wi-Fi



Radio Waves

Source:

Source: Oscillating electronic circuits

Radio broadcasting, television signals, remote-controlled toys, MRI